

Reproducing and Extending "From Attention to Activation": A Partial Replication Study of Attention Sinks and Activation Outliers

Tristan MARTIN, Ying JIN

April 2026

1 Introduction

While Transformer-based LLMs excel empirically, their mechanistic interpretability remain insufficiently understood [5]. The paper [6] investigates two particularly persistent anomalies observed in modern LLMs. The first anomaly is the systematic dominance of the first token in attention maps [9], where a large fraction of attention heads allocate a disproportionate amount of probability mass to the first key position. The second anomaly is the emergence of extreme activation outliers in a certain subset of hidden-state channels [4], with the most pronounced effect often appearing at the first token position.

This research is driven by mechanistic interpretability and system efficiency. First, addressing first-token attention dominance ensures that attention patterns reflect true semantic importance rather than training artifacts, thereby enhancing model transparency. Second, mitigating activation outliers narrows the dynamic range of hidden representations, which is essential for preserving precision during low-bit quantization.

The paper investigates the causal roots of these effects and proposes targeted interventions. The authors attribute the attention anomaly to the interaction between causal masking and canonical softmax normalization, where the "sum-to-one" constraint forces the first token to act as a privileged default destination when no strongly relevant key is present. To mitigate this, we introduce softmax-1 [6], defined as:

$$\text{softmax-1}(x_i) = \frac{\exp(x_i)}{1 + \sum_{j=1}^L \exp(x_j)}$$

By allowing the total attention mass to be less than one, the added constant serves as a "null" destination, preventing the first token from becoming a mandatory sink.

Simultaneously, they trace the emergence of activation outliers to the behavior of adaptive optimizers like Adam [7], which accumulate gradient moments in the parameter basis and can repeatedly amplify specific basis-aligned directions. To address this, they propose OrthoAdam, a variant that transforms gradients with a fixed orthogonal matrix Q before moment accumulation and maps the updates back to the parameter basis:

$$\bar{g}_t = Qg_t, \quad \bar{m}_t = \beta_1 \bar{m}_{t-1} + (1 - \beta_1) \bar{g}_t, \quad \theta_t = \theta_{t-1} - \eta Q^\top \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \right)$$

This design preserves the optimization benefits of adaptive methods while preventing the formation of outlier channels.

The methodology of this project adheres to the foundational logic of the original paper. We first adopt a mechanistic interpretability lens to diagnose the causal drivers behind two specific anomalies: attention sinks and activation outliers. Building upon these diagnostic insights, we implement the proposed solutions: softmax-1 and OrthoAdam to study if they can effectively mitigate these phenomena in a reduced setting.

2 Eliminating Candidate Causes

We conducted an initial analysis using a single forward pass across several pretrained model families (Pythia 14M–1.4B [2], GPT-2 124M–1.5B [8], and Llama-3-8B [1]) that cover a wide range of parameter scales and architectural configurations. By analyzing the results across these disparate models, we aim to eliminate candidate causes and identify the underlying mechanism.

As illustrated in Appendix 6 and 7 and table 1, while mainstream models such as GPT-2 and Llama exhibit both attention sinks and activation outliers, the Pythia series displays a distinct behavior. In smaller Pythia variants, attention dominance occurs but at first token. However, as the model scale exceeds 410M, the characteristic "first-token" attention sink begins to manifest. Regarding activation outliers, there is no clear relationship between model size and the emergence of extreme-magnitude channels. Most notably, Pythia-1.4B yields a surprisingly low Max Kurtosis of only 3.0, which means that this model doesn't have the activation outliers.

Table 1: Attention Sinks and Activation Outliers across Pretrained Model Families.

Model	Max Kurtosis	Sink Pos. (k)	Dom %	1st-Tok. Dom.	Act. Outliers
Pythia-14M	21	50	14.9%	No	Not extreme
Pythia-31M	38	50	16.0%	No	Not extreme
Pythia-70M	37	9	21.9%	No	Not extreme
Pythia-160M	18	9	47.3%	No	Not extreme
Pythia-410M	94	9,0	19.0, 43.0%	Yes	Not extreme
Pythia-1B	216	0	40.9%	Yes	Yes
Pythia-1.4B	3	0	40.2%	Yes	No
GPT-2 (124M)	689	0	65.0%	Yes	Yes
GPT-2-Medium (355M)	887	0	70.1%	Yes	Yes
GPT-2-Large (774M)	67	0	62.2%	Yes	Not extreme
GPT-2-XL (1.5B)	87	0	63.3%	Yes	Not extreme
Llama-3-8B	204	0	93.7%	Yes	Yes

We examine the structural commonalities that serve as primary candidate causes:

Feature	Pythia	GPT-2	Llama-3
Positional encoding	RoPE	learnt absolute	RoPE
Dedicated <bos> token	none	none	< begin_of_text >
Activation function	GeLU	GeLU	SiLU
Feature normalisation	LayerNorm	LayerNorm	RMSNorm

Table 2: Structural differences between the three public pretrained families

2.1 First token attention dominance

The presence of attention sinks across three types of model. This immediately eliminates several structural features as primary causes:

- **Design of Positional Encoding:** sink occurs in both RoPE-based architectures (Pythia, Llama-3) and those using learnt absolute encodings (GPT-2).
- **Dedicated BOS Token:** While Llama-3 utilizes a specific <|begin_of_text|> token, both GPT-2 and Pythia exhibit significant sinks despite the absence of a dedicated BOS token.
- **Activation and Normalization:** The phenomenon persists across different activation functions (SiLU vs. GeLU) and feature normalization techniques (RMSNorm vs. LayerNorm).

Emergence of Token 0 dominance We observe a distinct scaling-dependent shift in Pythia’s attention sinks. In smaller variants (14M–160M), attention sinks manifest at early positions such as $k = 50$ or $k = 9$

rather than the sequence start. This sink eventually stabilizes at token 0 once the scale reaches $410M$, though position 9 often retains a residual share of attention. We hypothesize that low-capacity models lack the structural maturity to internalize token 0 as a universal, functional 'offloading' coordinate. Instead, they rely on 'Semantic Anchors'—incidental high-frequency tokens or punctuation that appear consistently in early sequence positions. This is empirically validated in the Pythia setting: the attention sinks at $k = 9$ and $k = 50$ specifically correspond to the high-frequency tokens 'of' and 'each', respectively.

Positional Anchoring vs. Semantic Salience The architecture comparison already rules out the special BOS-token hypothesis. It remains possible that the sink is anchored by the semantic identity of typical English sentence starters. One could argue that capitalized, high-frequency opening words carry a learned salience that attracts redundant attention. To decouple token identity from positional coordinates, we replaced the tokens at first position with arbitrary mid-vocabulary IDs which the original paper does not perform. As shown in Table 3, swapping the token at the sink position leaves attention mass at that key essentially unchanged ($|\Delta B| < 0.6$ pp for every model). The sink remains fixed at the coordinate regardless of the token's meaning. This confirms the sink is purely positional, inherited by whatever token occupies the privileged index.

Model	Baseline B	Swap B	ΔB (pp)
GPT-2 (124M)	36.91 %	36.80 %	-0.10
GPT-2-Medium (355M)	42.17 %	41.94 %	-0.22

Table 3: **Swap Experiment.** Replacing the token at position 0 with an arbitrary ID results.

Length Dependence If the sink is driven by the causal mask, its intensity its influence should persist across varying context windows. As shown in Table 4, we evaluate the attention mass at $k = 0$ for the GPT-2 family across $T \in \{32, 128, 512, 1024\}$.

Model	$T=32$	$T=128$	$T=512$	$T=1024$
GPT-2 (124M)	52.3 % (17 $\times$)	43.2 % (55 $\times$)	36.9 % (189 $\times$)	35.8 % (366 $\times$)
GPT-2-Medium (355M)	58.8 % (19 $\times$)	48.8 % (62 $\times$)	42.2 % (216 $\times$)	41.8 % (429 $\times$)

Table 4: **Length dependence** mean attention mass% at key 0; parentheses give the ratio to the uniform $1/T$ baseline.) and is omitted there.

We see that while the absolute attention mass at key 0 decreases as T increases, the rate of decay is remarkably slow. For GPT-2-Medium, as the context window expands 32-fold (from $T = 32$ to $T = 1024$), the attention mass only regresses from 58.8% to 41.8%. More importantly, when measured against the uniform distribution baseline ($1/T$), the excess ratio grows monotonically and aggressively. At $T = 32$, the sink at $k = 0$ is approximately 19 $\times$ stronger than a uniform allocation; by $T = 1024$, this ratio surges to 429 $\times$. This resistance to dilution confirms that the sink is not a local semantic artifact but a structural constant of the architecture. The causal mask grants position 0 a unique topological privilege that the model increasingly exploits as the number of competing keys grows.

Softmax and Causal Masking Now for first token dominance, it only leaves us with existence of positional embedding (which we didn't train a model with/without positional embedding due to resources) and softmax and causal masking. Although we are unable to eliminate the pos embedding, the structural mechanics are more convinced: in a decoder-only Transformer, causal masking makes position 0 the only key visible to all queries. Consequently, the Softmax sum-to-one constraint forces it to act as a default 'dumping ground' for heads lacking strong semantic preferences."

To isolate this mechanism, we compared standard Softmax against Softmax-1 using synthetic constant and Gaussian logits. As shown in Appendix 5, standard Softmax (blue) is mathematically forced to assign 100% attention mass at query position 0, as it is the only visible key under the causal mask. Softmax-1 (red) introduces a non-zero null mass, which serves as a structural "escape hatch." This allows the model to

allocate mass to a virtual destination, effectively halving the artificial attention intensity at the sequence start. The consistent reduction in token-0 mass across both Constant and Gaussian regimes confirms that the sink is not driven by semantic features. Instead, it is a topological artifact: when a model is forbidden from attending to "nothing," it must default to the only universally visible coordinate.

2.2 Extreme Activation Outliers

By synthesizing cross-architecture evidence, we aim to isolate the drivers of extreme activation outliers (measured by Max Kurtosis).

Biases and Normalization

- **Feedforward Layer Biases:** While GPT-2 employs biases in all feedforward layers, the Llama and Pythia architectures omit them. Despite this, families with and without biases exhibit extreme outliers (e.g., Kurtosis of 887 for GPT-2 Medium and 204 for Llama-3-8B). This divergence suggests that layer biases are not a prerequisite for outlier formation.
- **Normalization Layer Scaling:** Both LayerNorm (GPT-2) and RMSNorm (Llama/Pythia) utilize per-channel scaling parameters. However, experimental evidence from "RMSNormSingle" configurations, where per-channel scaling is replaced by a single global scale, shows that outliers persist. Our cross-model comparison confirms this; for instance, Pythia-1.4B uses RMSNorm but achieves a Kurtosis of only 3.0 (near-Gaussian), while Pythia-1B reaches 216 using the same normalization scheme.

Decoupling Sinks from Activation Outliers It has been widely observed that activation outliers are spatially correlated with tokens receiving high attention scores, particularly at the sequence start [4, ?]. This has led to the common hypothesis that the attention sink mechanism is the primary driver behind the emergence of outlier channels. Our results provide a definitive counter-argument to this causal link as paper does:

- **The Pythia-1.4B Case:** This model possesses a strong first-token attention sink (40.2% dominance), yet it shows a complete absence of activation outliers (Max Kurtosis 3.0).
- **GPT-2 Scaling Variance:** In the GPT-2 family, the attention sink remains consistently high (> 60%) across all sizes. However, the Kurtosis drops by an order of magnitude from GPT-2 Medium (887) to GPT-2 XL (87).

Optimizer and learning rate With architectural and attention-based causes eliminated, will the coordinate-wise adaptive moments of the optimizer (e.g., Adam) be the driver?

Standard Adam [7] tracks the first (m_t) and second (v_t) moments of gradients independently for every parameter $\hat{\theta}_t = \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}$. Because Adam computes these updates in the same parameter basis as the model’s hidden representations, it can repeatedly amplify specific basis-aligned directions over millions of steps.

We hypothesize that this amplification manifests as a threshold-dependent phenomenon modulated by the learning rate η , following an interesting relationship between model size and activation outliers. In the GPT-2 series, outliers are most pronounced in small and medium-scale models. The learning rate in these models remains sufficiently high to catalyze Adam’s coordinate-wise amplification, while the expanded model width—relative to the smallest variants, provides the necessary numerical room for individual channels to drift. However, as the scale increases further into the billionaire-parameter regime, the mandatory reduction in learning rate required to maintain training stability effectively “cools down” the optimizer’s dynamics. We think this suppression prevents the cumulative moments from crossing the critical threshold of instability, thereby keeping the Kurtosis within a near-Gaussian range.

The near-Gaussian Kurtosis of 3.0 in Pythia-1.4B represents an extreme case of this stability. It remains to be determined whether such total suppression is a universal property of larger-scale models or an artifact of specific training configurations (e.g., specific β_2 values or epsilon constants). Nevertheless, the decoupling

of the attention sink from activation outliers in this model reinforces the hypothesis that outliers are a numerical byproduct of the optimization trajectory rather than a structural necessity.

Preventing Outliers via Basis Rotation OrthoAdam prevents outlier formation by ensuring that the basis for adaptive scaling is non-aligned with the feature basis. Mathematically, it replaces the standard coordinate-wise update $\Delta\theta = \text{Adam}(g)$ with:

$$\Delta\theta = Q^\top \text{Adam}(Qg)$$

Specifically, it applies a fixed, randomly sampled orthogonal matrix Q to the gradients. After the update is computed in this rotated space, it is mapped back to the original parameter basis via $Q^\top$ for the weight update. Thus the L_2 norm of the update remains invariant ($\|\Delta\theta\|_2 = \|\bar{u}\|_2$), preserving Adam’s convergence energy. However, the L_∞ norm (maximum outlier value) is significantly reduced because the transformation distributes the update across all dimensions, effectively reducing potential outlier components across the entire feature vector. Theoretically, OrthoAdam enables the model to maintain the fast convergence of adaptive methods while producing the stable, near-Gaussian activation profiles like SGD.

3 Experiment Setup

To study the identified structural artifacts, we established a controlled experimental setup using a $\sim 60\text{M}$ -parameter GPT-2 architecture, matching the smallest scale used in the original paper. To systematically evaluate the proposed interventions, we conduct an ablation study using a 60M-parameter GPT-2 architecture. We pretrain four model variants from scratch on the C4 dataset to isolate the effects of structural and numerical modifications: (1) a baseline GPT-2, (2) a variant isolating the softmax-1 intervention, (3) a variant isolating the OrthoAdam optimizer, and (4) a unified model combining both. Testing the combined variant was crucial to ensure that simultaneous interventions in attention routing and optimization trajectories do not induce antagonistic dynamics, but rather yield a fully denoised, quantization-friendly computational graph.

Methodology and Architectural Interventions

The interventions were implemented strictly from first principles, ensuring we did not rely on unverified third-party code.

- **Softmax-1:** We engineered a custom, numerically stable log-sum-exp formulation of the softmax-1 function. Through targeted model surgery, we injected this function into the Hugging Face `GPT2Attention` blocks, forcing the model to allow attention mass to drop below 1.
- **OrthoAdam:** We implemented the OrthoAdam optimizer as a custom PyTorch subclass. To bypass the prohibitive memory footprint of generating full $D \times D$ orthogonal matrices for large parameter tensors, we utilized axis-wise Kronecker rotations driven by Haar-distributed random matrices, applying an upper bound on the rotated dimension (e.g., leaving the massive vocabulary axis unrotated).
- **Ablation Alignments:** To perfectly match the paper’s specific test conditions across all variants, we systematically stripped all MLP layer biases and replaced standard LayerNorms with a custom single-scalar RMSNorm (`RMSNorm-S`).

Due to local compute constraints (utilizing a single P100 GPU), we adapted the training hyperparameters slightly: we reduced the effective batch size and trained in strictly FP32 precision, as FP16 optimization exhibited numerical divergence at the paper’s prescribed learning rates.

The Interpretability Pipeline

To evaluate these interventions, we built an analysis pipeline using custom PyTorch “forward hooks” to passively capture hidden states. Because storing all activations in GPU memory at once is prohibitive, we used Welford-style running accumulators to compute our statistics dynamically. This allowed us to efficiently calculate the exact per-layer kurtosis (a statistical measure of extreme outliers) and maximum absolute activations, precisely mapping where structural artifacts emerge.

4 Analysis of the Results

To systematically evaluate the efficacy of the architectural interventions, we analyze both the attention mass distribution and the geometric stability of the residual stream. Rather than immediately inspecting isolated layers, we first establish the macroscopic impact of our interventions. Figure 1 presents the aggregated quantitative results of our replication across all four architectural variants. We report three primary metrics: the average percentage of attention mass allocated to the first token, the mean kurtosis of the hidden states (serving as a proxy for heavy-tailed representations), and the global maximum absolute activation.

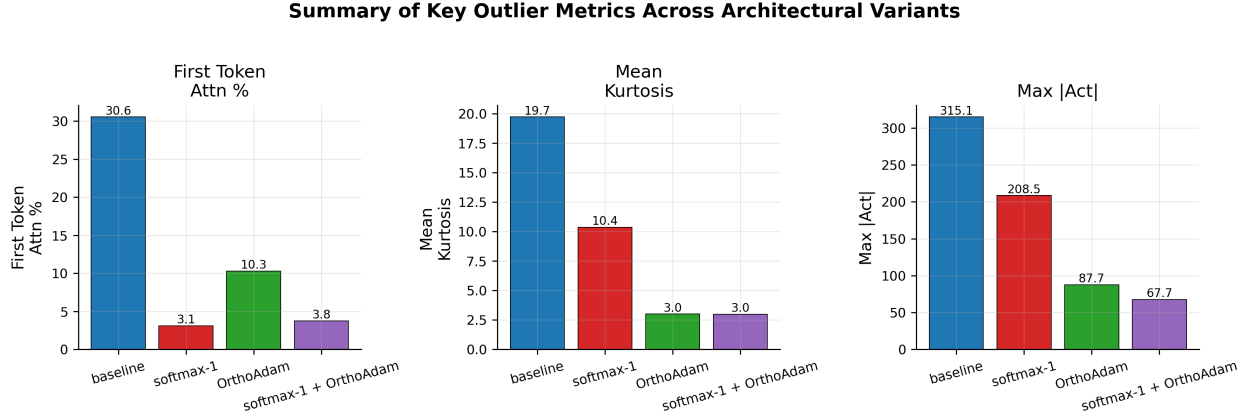


Figure 1: Summary of key outlier metrics across the four model variants. Softmax-1 effectively eliminates the first-token attention sink, while OrthoAdam successfully normalizes the representation space by minimizing mean kurtosis and bounding the maximum absolute activation.

4.1 Mitigating Attention Sinks with Softmax-1

Consistent with previous findings, our results confirm that the canonical softmax function induces a structural “attention sink.” As shown in Figure 1, the baseline model allocates 30.6% of its attention mass to the first token. In contrast, the softmax-1 intervention effectively mitigates this effect, reducing first-token attention to 3.1% (isolated) and 3.8% (unified). Without this intervention, even with the OrthoAdam optimizer, the model continues to exhibit an inflated attention rate (10.3%), confirming that the sink is a structural issue inherent to the softmax operation rather than the optimization trajectory.

As illustrated in Figure 2, this dominance in the baseline model fluctuates heavily across network depth, reaching peaks of nearly 60% in the middle layers before the final unembedding projection. Softmax-1 successfully flattens this trajectory, proving that relaxing the sum-to-one constraint effectively sanitizes the attention circuitry. For a qualitative visual verification of this phenomenon, we refer the reader to Appendix 8, where the attention heatmaps clearly demonstrate the disappearance of the characteristic first-column sink.

Interestingly, while OrthoAdam is primarily a numerical intervention, it also exhibits a stabilizing effect on the attention sink trajectory in the deeper layers (Layers 4–11). Unlike the baseline, which suffers from recurrent attention spikes throughout the network, OrthoAdam prevents the exponential growth of activation outliers. By maintaining a well-behaved representation space, it indirectly mitigates the feedback loop between numerical instability and attention concentration. However, it fails to suppress the sink in the initial layers (Layers 1–2), where the structural constraints of the canonical softmax remain the dominant driver of the artifact. Only **Softmax-1** achieves a universal flattening across all depths by addressing the root structural cause.

4.2 Stabilizing Activation Distributions via OrthoAdam

Concurrently, we evaluate the emergence of activation outliers by measuring the kurtosis (κ) of the hidden states, which serves as a proxy for heavy-tailed, outlier-driven distributions in the representation space.

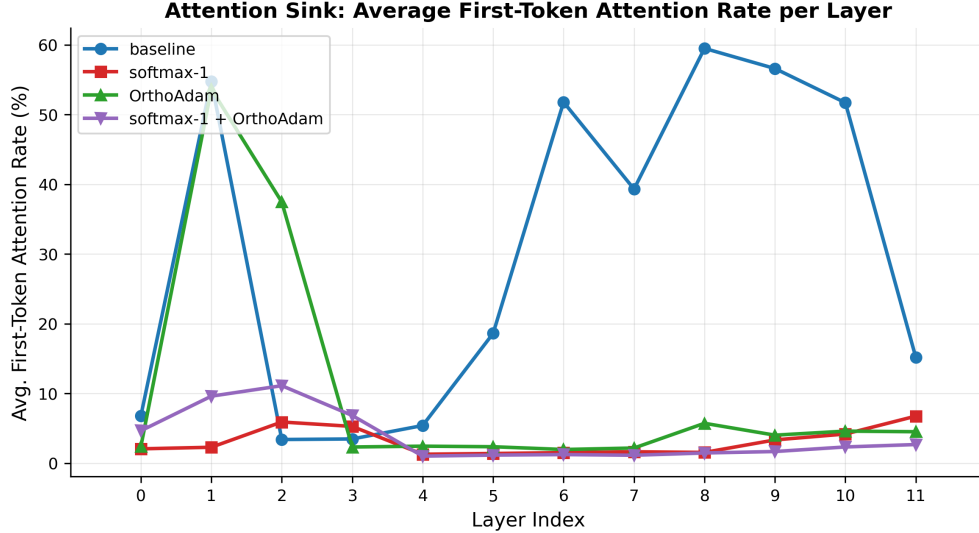


Figure 2: Layer-wise progression of first-token attention dominance. Softmax-1 systematically suppresses the attention sink across all depths.

Mechanistically, adaptive optimizers tend to artificially inflate specific feature channels, permanently warping the residual stream.

Our empirical results confirm this hypothesis. The baseline model exhibits an extreme kurtosis on the first token ($\kappa = 284.0$), indicating Extreme Activation Outliers. The introduction of OrthoAdam systematically resolves this. By rotating the gradient updates via orthogonal matrices, OrthoAdam prevents the parameter basis from aligning with the optimizer’s historical momentum. Consequently, both runs utilizing OrthoAdam exhibit a remarkable drop in first-token kurtosis, converging perfectly to the Gaussian reference of $\kappa = 3.0$. As shown in Figure 3, while the baseline model’s kurtosis explodes exponentially as a function of depth, the OrthoAdam variants seamlessly hug the Gaussian reference line ($\kappa = 3$) throughout the entirety of the forward pass. This confirms that OrthoAdam preserves a mathematically well-behaved representation space, devoid of the extreme magnitude distortions that traditionally confound interpretability probes and low-bit quantization efforts.

Beyond the statistical shape of the distribution, the absolute magnitude of these outlier activations poses a distinct and critical threat to mechanistic interpretability. Modern interpretability techniques, particularly Sparse Autoencoders (SAEs) and linear probes, rely on the geometric variance of the residual stream to disentangle polysemantic neurons and map linear representations of concepts. When a subset of channels exhibits extreme activation magnitudes, they disproportionately dominate the L_2 norm of the hidden state. This effectively blinds feature extraction algorithms, causing them to allocate dictionary features to optimizer-induced artifacts while “washing out” smaller, genuinely semantic activation patterns.

As illustrated in Figure 4, we observe an exponential explosion in the maximum absolute activation ($|a|$) of the first token in the baseline model, particularly emerging in the middle layers of the network. While the isolated softmax-1 intervention provides partial mitigation by reducing the arbitrary routing of attention mass, the application of OrthoAdam fundamentally constrains these magnitudes. Crucially, the unified **softmax-1 + OrthoAdam** variant achieves the most profound stabilization, decisively compressing the dynamic range of the hidden states across all depths. By rigorously bounding these maximum activations, we ensure that the residual stream’s geometry remains dense and semantically expressive, rather than being warped by singular, artifactual feature spikes.

These results were validated across a wide range of hyperparameters of learning rate and beta value which are hyperparameter of the Adam optimizer.

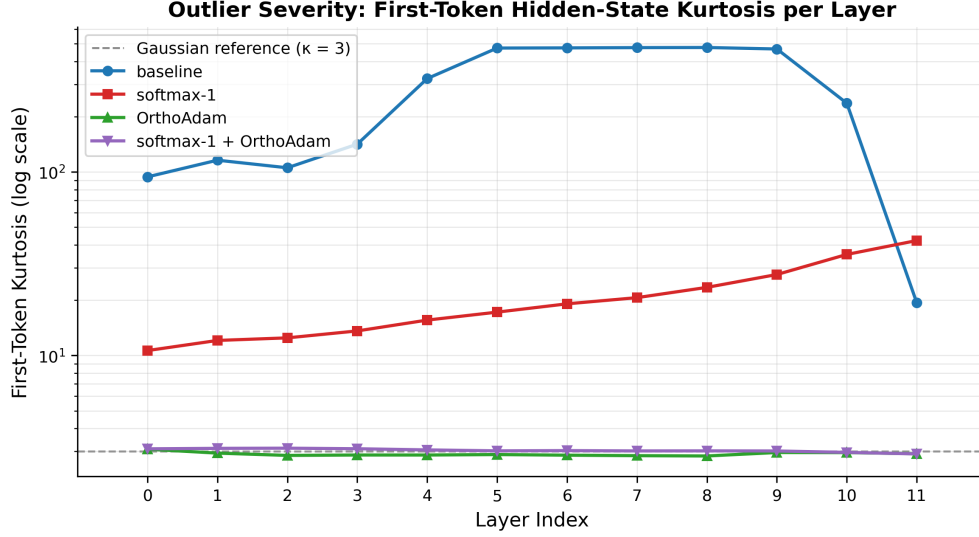


Figure 3: Layer-wise progression of first-token hidden-state kurtosis (log scale). OrthoAdam tightly bounds the kurtosis to the Gaussian reference ($\kappa \approx 3$), eliminating extreme outlier emergence.

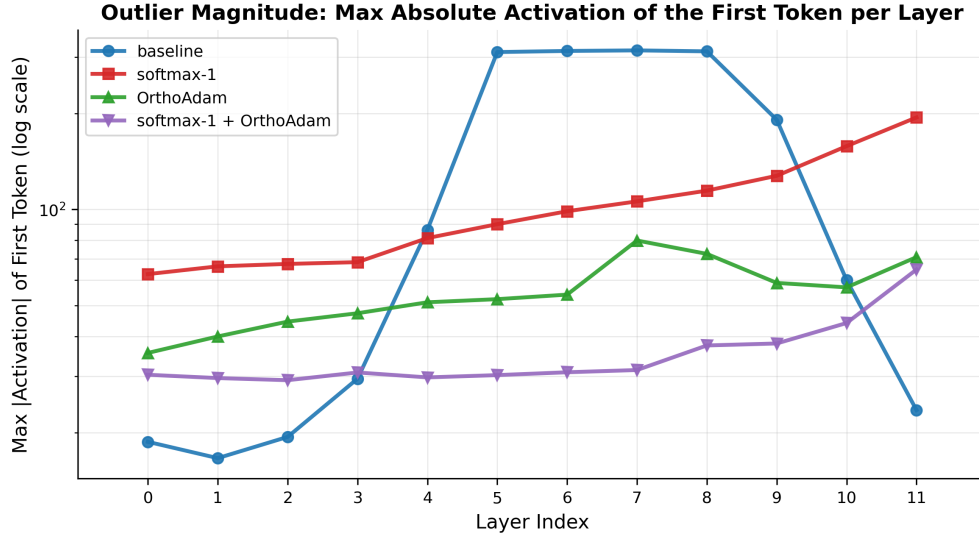


Figure 4: Layer-wise progression of the maximum absolute activation ($|a|$) for the first token (log scale). The baseline model exhibits an exponential explosion in activation magnitude, which is systematically compressed and stabilized by the OrthoAdam interventions.

5 Conclusion

This project successfully replicates and validates the structural and optimization interventions proposed by Kaul et al. for addressing pervasive artifacts in auto-regressive Transformers. Our empirical results lead to the following key conclusions:

- **Mitigating Attention Sinks with Softmax-1** We confirm that the attention sink is a mathematical byproduct of the canonical sum-to-one constraint. By introducing softmax-1, we allow the model to dynamically "discard" irrelevant attention mass into a zero-vector slot. This effectively reduces first-token attention from a disproportionate 30.6% to a semantically grounded 3.1%, ensuring that the model's focus is dictated by logic rather than structural necessity.
- **Numerical Stability via OrthoAdam:** While softmax-1 addresses structural artifacts in attention, OrthoAdam ensures numerical consistency within the model's internal states. By preventing parameters from aligning with the optimizer's historical momentum, it constrains the distribution of the representation space. Our results demonstrate that kurtosis is maintained at the Gaussian-ideal of $\kappa \approx 3.0$, effectively eliminating the extreme activation outliers ($\kappa = 284.0$) that characterize the baseline model's deep residual streams.
- **The Synergy of Joint Interventions:** A crucial insight from our ablation study is the complementary nature of these techniques. While OrthoAdam alone stabilizes deep-layer trajectories, it cannot resolve the initial-layer attention sink. Conversely, softmax-1 fixes the sink but does not fully suppress activation explosions. Only the unified application of both interventions achieves a clean representation space that is stable across all depths and positions.

This replication underscores a foundational principle of Explainable AI and Mechanistic Interpretability: before we can accurately map the true computational circuitry of an LLM, we must first sanitize its representation space. Interventions like softmax-1 and OrthoAdam do not merely optimize training dynamics or facilitate robust low-bit quantization; they act as critical denoising mechanisms. By clearing the fog of optimization artifacts, these techniques ensure that the network's geometry is dictated solely by authentic semantic features, paving the way for more reliable feature extraction and deeper mechanistic transparency.

6 Future Work

While this replication confirms the efficacy of softmax-1 and OrthoAdam at the current scale, future research should explore the benefits of these interventions in real practice. With softmax-1 that resolves the attention sink, it is worth evaluating the impact on extreme context window extension and the efficiency of KV-cache compression in long-range dependency modeling [?]. The near-Gaussian representation space maintained by OrthoAdam offers a superior foundation for both low-bit quantization and mechanistic interpretability.

Furthermore, Modern mechanistic interpretability relies heavily on SAEs to disentangle features. However, outliers often "blind" these models. By using a representation space stabilized by OrthoAdam, future research could investigate whether SAEs can achieve higher reconstruction fidelity and better feature sparsity, as suggested by the foundational work on feature steering by [3].

7 Usage of LLMs

To be transparent about our work, we leveraged LLMs tools for software engineering and manuscript preparation. In the engineering phase, LLMs assisted with PyTorch boilerplate and data pipelines. However, all complex model modifications were subject to rigorous, human-led testing to resolve potential logic errors. For the final report, LLMs were used to refine the academic tone and manage LaTeX formatting.

References

- [1] Meta AI. The llama 3 herd of models, 2024.

- [2] Stella Biderman et al. Pythia: A suite for analyzing large language models across training and scaling. *ICML*, 2023.
- [3] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.
- [4] Tim Dettmers et al. LLM.int8(): 8-bit matrix multiplication for transformers at scale. In *NeurIPS*, 2022.
- [5] Nelson Elhage et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021.
- [6] Kushal Kaul et al. From attention to activation: A mechanistic view on sinks and outliers. *arXiv preprint*, 2025.
- [7] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- [8] Alec Radford et al. Language models are unsupervised multitask learners. 2019.
- [9] Guangxuan Xiao et al. Efficient streaming language models with attention sinks. In *ICLR*, 2023.

8 Appendix

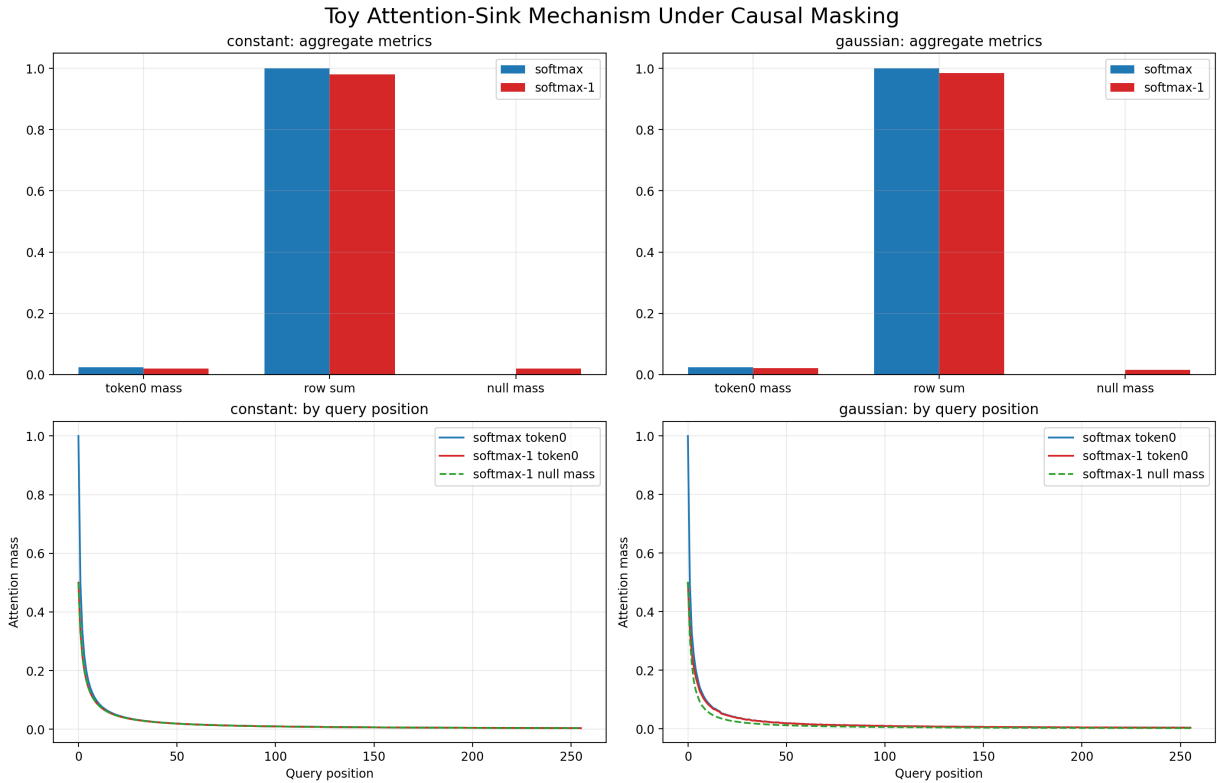


Figure 5: Toy Attention Sink mechanism under causal masking

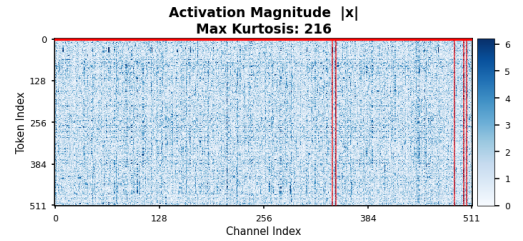
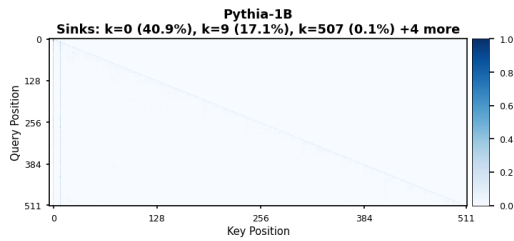
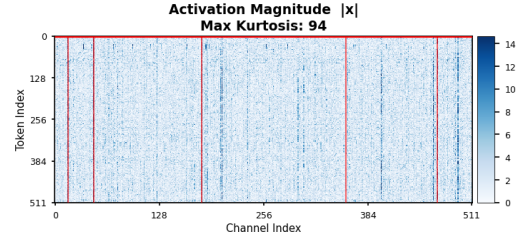
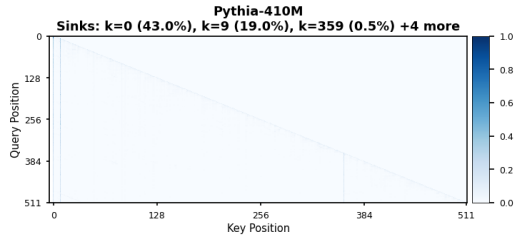
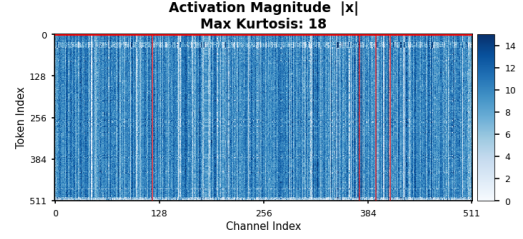
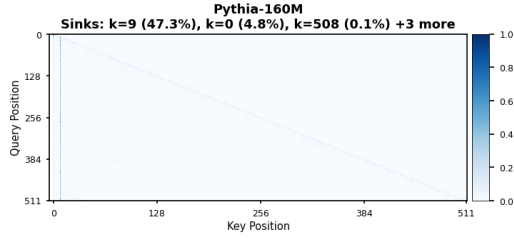
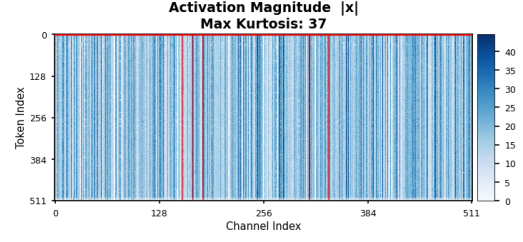
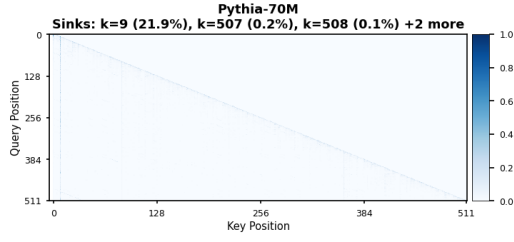
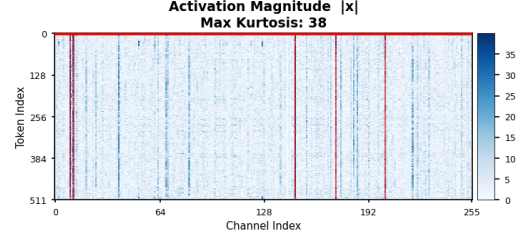
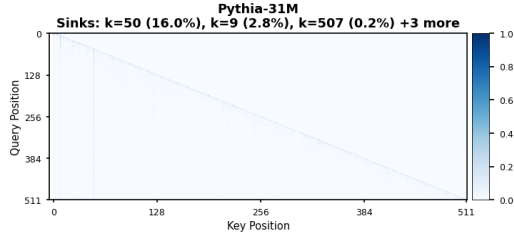
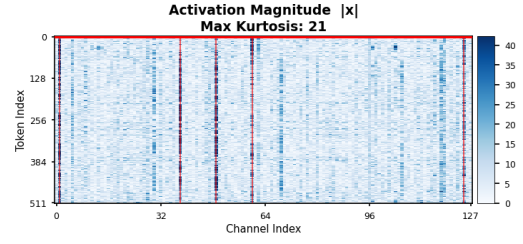
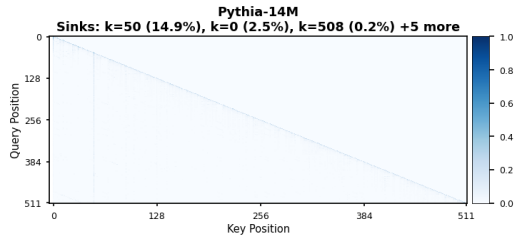


Figure 6: Pythia family exhibits both the attention sink but not key 0 and extreme outlier channels in the last-layer hidden state

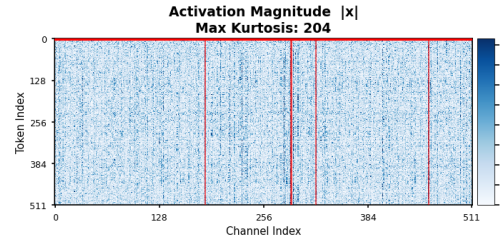
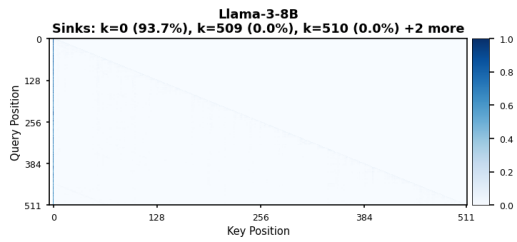
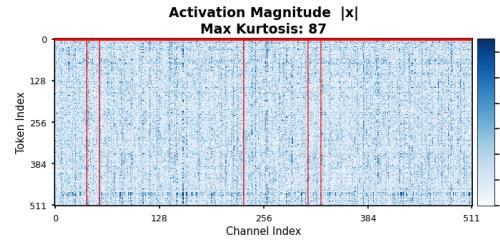
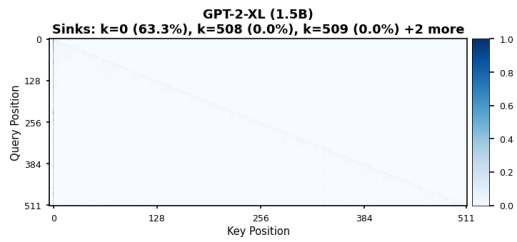
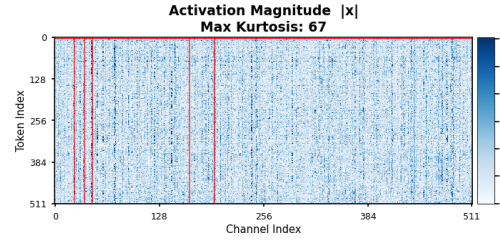
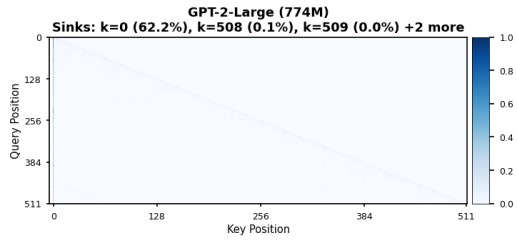
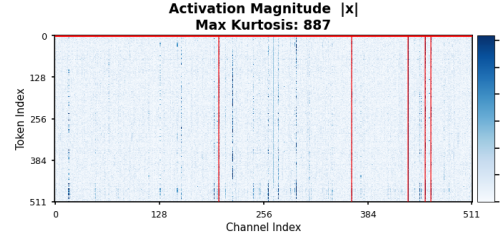
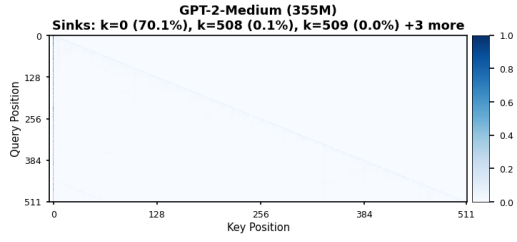
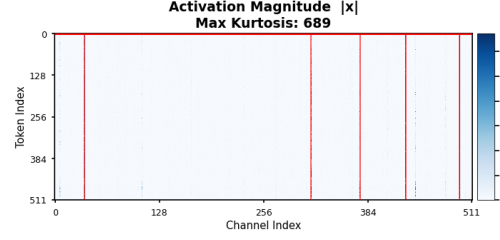
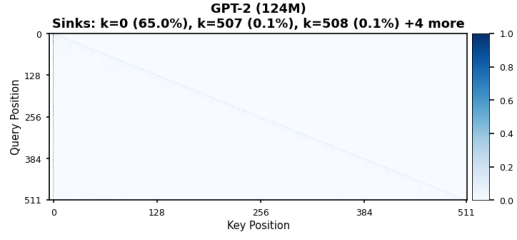
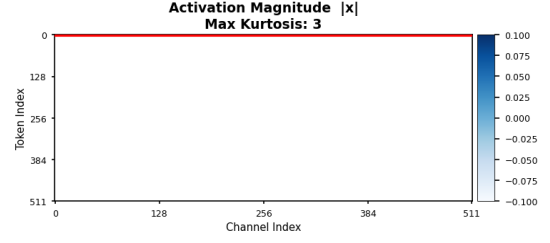
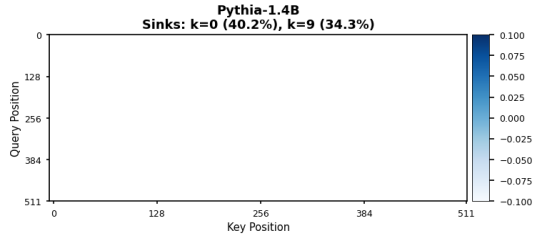


Figure 7: Pythia-1.4B, GPT-2 series, Llama-3 8B exhibits both the attention sink on key 0 and extreme outlier channels in the last-layer hidden state

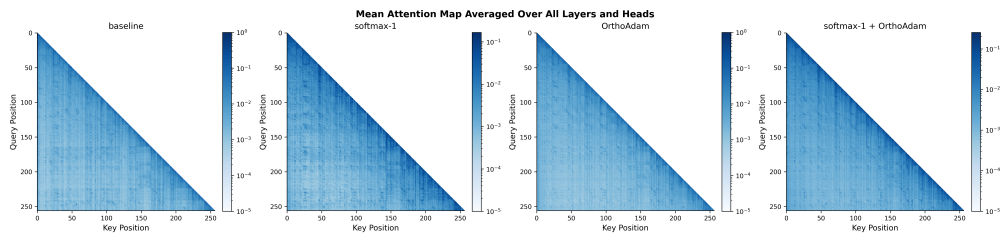


Figure 8: Mean Attention Map Averaged over all layers and heads